# MATHEMATICS

## SECTION – I

## Single Correct Choice Type

**1. (b)** Option (b) is correct because in this case $a > 0,\ b < 0$ (let $b = -c$)

$$\Rightarrow \quad -cx^2 + ay^2 = -ac \quad \Rightarrow \quad \frac{x^2}{a} - \frac{y^2}{c} = 1$$

**2. (a)** $f : R \to R,\ f(x) = \dfrac{x^2 + bx + 1}{x^2 + 2x + b}$, Range must be of the form $\left[\dfrac{1}{a}, a\right]$

So numerator and denominator never be zero.

$\Rightarrow \ D < 0 \ \Rightarrow \ 4 - 4b < 0 \ \Rightarrow \ b > 1$ (for denominator)

$\Rightarrow \ b^2 < 4 \ \Rightarrow \ -2 < b < 2$ (for numerator)

$$y = \frac{x^2 + bx + 1}{x^2 + 2x + b} \quad \Rightarrow \quad (y-1)x^2 + (2y - b)x + by - 1 = 0$$

If we make its discriminant zero, then its equation has root of form $a, \dfrac{1}{a}$.

$$\Rightarrow \quad 4(1-b)y^2 + 4y + b^2 - 4 = 0 \ \Rightarrow \quad \frac{b^2 - 4}{4(1-b)} = 1 \quad \Rightarrow \quad b = -2 \pm 2\sqrt{3}$$

**3. (b)** $(\cos y - \sin y)\dfrac{dy}{dx} + (\cos y + \sin y)e^{-x} = e^{e^{-x}}$

$\cos y + \sin y = u$

$$\Rightarrow \quad \frac{du}{dx} + e^{-x}u = e^{e^{-x}} \quad \Rightarrow \quad ue^{-e^{-x}} = x$$

$$\Rightarrow \quad e^{-e^{-t}} = t \qquad \text{(putting } (t, 0)\text{)}$$

$$\Rightarrow \quad t.e^{e^{-t}} = 1$$

**4. (c)** Multiply by $y$, $z$ and $x$ in rows 1, 2 and 3 respectively and then take common $y$, $z$ and $x$ from column 1, 2 and 3 respectively, then

$$\begin{vmatrix} y^3+1 & y^3 & y^3 \\ z^3 & z^3+1 & z^3 \\ x^3 & x^3 & x^3+1 \end{vmatrix} = 11$$

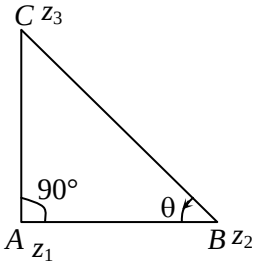
$$\Rightarrow \begin{vmatrix} 1 & 0 & y^3 \\ -1 & 1 & z^3 \\ 0 & -1 & x^3+1 \end{vmatrix} = 11 \qquad \left(C_1 \to C_1 - C_2 \text{ and } C_2 \to C_2 - C_3\right)$$

$$\Rightarrow \ 1\left(x^3 + 1 + z^3\right) + y^3(1) = 11 \ \Rightarrow \ x^3 + y^2 + z^3 = 10$$

So solution are (1, 1, 2), (1, 2, 1) or (2, 1, 1)

## SECTION – II
### Multiple Correct Choice Type

**5.** **(a), (b)**




$$\frac{W_2}{|W_2|} = \frac{W_1}{|W_1|}e^{-i\theta}$$

$$\Rightarrow \quad \frac{W_2}{W_1} = \frac{|W_2|}{|W_1|}e^{-i\theta} = 1 - i\tan\theta$$

$$W_1\overline{W_2} = \frac{W_1}{W_2}|W_2|^2 = |W_1||W_2|e^{i\theta} = \left(|W_1|^2 + i|W_1||W_2|\sin\theta\right)$$

**6.** **(a), (d)**

$$(1+x+x^2)^n = a_0 + a_1x + a_2x^2 + \ldots + a_{2n}x^{2n} \qquad \ldots(1)$$

$$\Rightarrow \quad a_0 + a_2 + a_4 \ldots = \frac{3^n+1}{2}$$

$$a_1 + a_3 + a_5 \ldots = \frac{3^n-1}{2}$$

Put $x = i$ in equation (1) where $i = \sqrt{-1}$

If $n = 4m+1$

$\Rightarrow \quad a_1 - a_3 + a_5 \ldots = 1$

$a_0 - a_2 + a_4 \ldots = 0$

So, $a_1 + a_5 + a_9 \ldots = \frac{3^n+1}{4} = S_2$

$$a_3 + a_7 + a_{11} \ldots = \frac{3^n-3}{4} = S_4$$

Similarly if $n = 4m+3$

Then, $S_2 = \frac{3^n-3}{4}$ and $S_4 = \frac{3^n+1}{4}$

**7.** **(c), (d)**

$$(13x-1)^2 = 36$$

$$13x - 1 = \pm 6$$

$$x = \frac{7}{13}$$

$$x = -\frac{5}{13}$$

$$\Rightarrow \quad \cos\alpha = -\frac{5}{13}; \quad \sin\alpha = \pm\frac{12}{13}$$

$$\sin 2\alpha = \pm\frac{120}{169}$$

**8.** **(a), (c), (d)**

$$\cos^2(\pi x) - \sin^2(\pi y) = \frac{1}{2}$$

$$\Rightarrow \quad \cos\{\pi(x+y)\}\cos\{\pi(x-y)\} = \frac{1}{2}$$

$$\Rightarrow \quad \cos\{\pi(x+y)\}\cos\left(\frac{\pi}{3}\right) = \frac{1}{2} \qquad \left(\because x - y = \frac{1}{3}\right)$$

$$\Rightarrow \quad \cos\pi\{x+y\} = 1 \;\Rightarrow\; \pi(x+y) = 2n\pi \;\Rightarrow\; x + y = 2n,\ n \in I$$

Now, $x + y = 2n$ and $x - y = \frac{1}{3}$

$$\Rightarrow \quad x = n + \frac{1}{6},\ y = n - \frac{1}{6}$$

$$\Rightarrow \quad (x,\ y) = \left(n + \frac{1}{6},\ n - \frac{1}{6}\right)$$

(a), (c), (d) correspond $n = 1, -1, 2$ respectively.

## SECTION – III
## Reasoning Type

**9.** **(b)** As both foci lies on opposite side of directrix the curve must be hyperbola and statement 2 is true but does not explain the first one.

**10.** **(d)** $\lim\limits_{x\to 0}\left(\frac{1+2x}{1+3x}\right)^{\frac{1}{x^2}} e^{\frac{1}{x}} = \lim\limits_{x\to 0} e^{\frac{\ln(1+2x)-\ln(1+3x)+x}{x^2}}$ (using $L$' hospital rule)

$= e^{5/2}$

**11.** **(a)** $\lim\limits_{t\to 0}\int_0^{2\pi}\left|\frac{\sin(x+t)-\sin x}{t}\right| dx = \int_0^{2\pi} |\cos x|\, dx = 4$

**12.** **(a)** As we know that $\left(a_1^2 + a_2^2 ..... + a_n^2\right)\left(b_1^2 + b_2^2 ..... + b_n^2\right) \geq \left(a_1 b_1 + a_2 b_2 + ..... + a_n b_n\right)^2$

Now, put $b_1 = b_2 = b_3 ..... b_n = 1$

Now let $a_r = f\left(\frac{r}{n}\right)$

$$\Rightarrow \quad \sum_{r=1}^{n} f^2\left(\frac{r}{n}\right)\frac{1}{n} \geq \left(\sum_{r=1}^{n}\frac{f(r/n)}{n}\right)^2 \;\Rightarrow\; \lim_{n\to\infty}\sum_{r=1}^{n} f^2\left(\frac{r}{n}\right)\frac{1}{n} \geq \left(\lim_{n\to\infty}\sum_{r=1}^{n}\frac{f(r/n)}{n}\right)^2$$

$$\Rightarrow \quad \int_0^1 f^2(x)dx \geq \left(\int_0^1 f(x)dx\right)^2$$

## SECTION–IV
## Comprehension Type

### Passage-I

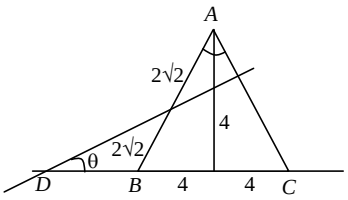
**Sol.:**

**13. (d)** Area of quadrilateral will be

$$= 16 - \frac{1}{2} 2\sqrt{2}.2\sqrt{2} \tan(45 - \theta)$$

$$= 16 - 4\left(\frac{1 - \tan\theta}{1 + \tan\theta}\right)$$

$$= \frac{4(4 + 4\tan\theta - 1 + \tan\theta)}{1 + \tan\theta} = 4\left(\frac{3 + 5\tan\theta}{1 + \tan\theta}\right)$$




**14. (b)** Length of portion of straight line inside the triangle = $2\sqrt{2}\sec\left(\frac{\pi}{4} - \theta\right)$.

**15. (c)** $x^3 + y^3 + (-1)^3 - 3(-1)(x)(y) = 0$

$\Rightarrow (x + y - 1)[(x - y)^2 + (x + 1)^2 + (y + 1)^2] = 0$

$\Rightarrow$ one point (–1, –1) and two points satisfies the equation $x + y = 1$.

So, three isosceles triangle are possible such that one side length 8 and two angles are equal.

### Passage-II

**16. (d)** $\int_0^{\pi} g(x)\,dx =$ Imaginary part of $\sum_{r=1}^{n} \int_0^{\pi} b_r e^{irx}\,dx$

$$= \sum_{r=1}^{n} \frac{2b_r}{r} \qquad (r \text{ is odd integer})$$

$$\lim_{n\to\infty} \int_0^{\pi} g(x)\,dx = 2\left(\frac{b_1}{1} + \frac{b_3}{3} + \frac{b_5}{5} + \ldots\right) = 2\left(\frac{1}{1.2} + \frac{1}{3.4} + \frac{1}{5.6} + \ldots\right)$$

$$= 2\left(\frac{1}{1} - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \ldots\right) = 2\ln 2$$

**17. (a)** $\int_0^{2\pi} f(x)\cos kx\,dx = \sum_{r=0}^{n} \int_0^{2\pi} a_r \cos rx \cos kx\,dx$

$$= \frac{1}{2}\sum_{r=0}^{n} \int_0^{2\pi} a_r(\cos(r + k)x + \cos(r - k)x)\,dx$$

$$\frac{1}{2}\sum_{r=0}^{n}\left[a_r\left(\frac{\sin(r + k)x}{(r + k)} + \frac{\sin(r - k)x}{(r - k)}(r \neq k)\right)\right]_0^{2\pi} + \frac{1}{2}a_k \int_0^{2\pi} (\cos 2kx + 1)\,dx = \pi a_k$$

$$\Rightarrow \qquad \sum_{r=0}^{n} \int_0^{2\pi} f(x)\cos kx\,dx = \pi(a_0 + a_1 + a_2 + \ldots a_n = \pi f(0)$$

**18.** **(d)** $\int_0^{2\pi} b_r \sin rx \cos kx \, dx = \frac{1}{2} b_r \int_0^{2\pi} (\sin(r+k)x + \sin(r-k)x) \, dx = 0$

$\sum_{k=0}^{n} \int_0^{2\pi} b_r \sin rx \cos kx \, dx = 0$

## SECTION–V
## Matrix–Match Type

**1.** **(A) – 1, 4; (B) – 1, 3; (C) – 1, 3; (D) – 2, 4, 5**

**(A)** $f(x) = (x+1)^2 + x - [\sqrt{(x+1)^2 + (x+1)}]^2$

$(x+1)^2 < (x+1)(x+2) < (x+2)^2$ (as $x \in N$)

$(x+1) < \sqrt{(x+1)(x+2)} < (x+2)$

$\Rightarrow \quad f(x) = x$

So, $f(x)$ is one-one and into function

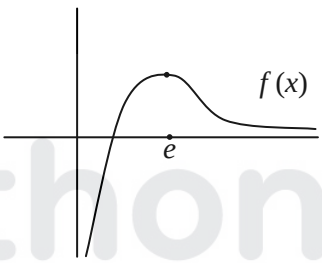
**(B)** $\frac{\log g(x)}{g(x)} = x$

$\Rightarrow \quad y = g(x)$ is inverse of $y = f(x)$

so its enough to investigate

$f : (0, 1) \to (-\infty, 0), \; y = f(x)$

which is one-one onto as




$f'(x) = \frac{1 - \log x}{x^2} = \frac{\log\left(\frac{e}{x}\right)}{x^2} > 0$

**(C)** $f : R \to R$ be continuous function

Such that $|f(x) - f(y)| \geq |x - y|$

$\Rightarrow \quad f(x) = f(y)$

If $x = y$

So $f$ is one-one

Since $f(x)$ is continuous for all $x \in R$ and it is not bounded function.

$\Rightarrow \quad f(x)$ is onto function

**(D)** $f(x) = \int_0^x \frac{1}{1+t^2} dt + \int_0^{1/x} \frac{1}{1+t^2} dt$

$f(x) = \tan^{-1} x + \tan^{-1}\left(\frac{1}{x}\right)$

$f(x) = \frac{\pi}{2}$ for $x > 0$

$\frac{-\pi}{2}$ $x < 0$

So $f(x)$ is many one and into

**2. (A) – 1, 2; (B) – 2, 3, 4 ; (C) – 4, 5; (D) – 5**

**(A)** $|z_1|=|z_2|=|z_3|=1$

$z_1^3+z_2^3+z_3^3-3z_1z_2z_3=-4z_1z_3z_3$

or $(z_1+z_2+z_3)\left((z_1+z_2+z_3)^2-3(z_1z_2z_3)(\bar{z}_1+\bar{z}_2+\bar{z}_3)\right)=-4z_1z_2z_3$

or $(z_1+z_2+z_3)^3=z_1z_2z_3(3|z_1+z_2+z_3|^2-4)$

or $|z_1+z_2+z_3|^3=\left|3|z_1+z_2+z_3|^2-4\right|$ or $k^3=|3k^2-4|$

If $k>\dfrac{2}{\sqrt{3}}$

$k^3=3k^2-4$

$k^3-3k^2+4=0 \quad \Rightarrow \quad (k-2)^2(k+1)=0 \Rightarrow k=2$

If $0<k<\dfrac{2}{\sqrt{3}}$ $\quad k^3+3k^2-4=0 \Rightarrow (k-1)(k+2)^2=0 \Rightarrow k=1$

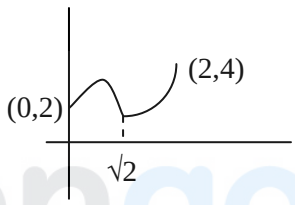
**(B)** Let $|1-z|=t$

So $0\le t\le 2$ or $z+\bar{z}=2-t^2$

Let $P=|1-z|+|1+z^2|$

$P=t+|z||z+\bar{z}|$

$P=t+|2-t^2|,\ 0\le t\le 2$




**(C)** $|1-\cos\theta-i\sin\theta|=|2\sin^2\dfrac{\theta}{2}-2i\sin\dfrac{\theta}{2}\cos\dfrac{\theta}{2}|=2|\sin\dfrac{\theta}{2}|$

So $k=\pm 2$

**(D)** $f(x)=x^4+ax^3+bx^2+cx+d$

As $\alpha, \beta, \gamma, \delta$ are real roots of equation $f(x)=0$

$\Rightarrow \quad f(x)=(x-\alpha)(x-\beta)(x-\gamma)(x-\delta)$

Given that $|f(i)|=1$

$\Rightarrow \quad (1+\alpha^2)(1+\beta^2)(1+\gamma^2)(1+\delta^2)=1$

It is only possible if $\alpha=\beta=\gamma=\delta=0$

So $a=b=c=d=0$